Funktion aufstellen Lösungen · mehr

1.	**$c = 30$; $b = 0$; $f(x) = -0{,}007x^2 + 30$**	$f(x) = ax^2 + bx + c$; $f(0) = h(0) = 30 \Rightarrow c = 30$; $f'(0) = h'(0) = 0 \Rightarrow b = 0$; $g(40) = 14{,}4 \Rightarrow f(40) = 18{,}8 \Rightarrow 1600a + 30 = 18{,}8 \Rightarrow a = -0{,}007 \Rightarrow f(x) = -0{,}007x^2 + 30$
2.	**$g'(t) = 3at^2 + 2bt$; $48a + 8b = 144$; $g(t) = -3t^3 + 36t^2$**	$g'(t) = 3at^2 + 2bt$; $g(4) = 384 \Rightarrow 64a + 16b = 384$; $g'(4) = 144 \Rightarrow 48a + 8b = 144$; $\Rightarrow a = -3$, $b = 36 \Rightarrow g(t) = -3t^3 + 36t^2$
3.	**$q(x) = ax^2 + bx$ ($q(0) = 0$); $6a + b = 0$; $q(x) = -0{,}3x^2 + 1{,}8x$**	$q(x) = ax^2 + bx$ ($q(0) = 0$); $q(3) = 2{,}7 \Rightarrow 9a + 3b = 2{,}7$; $q'(3) = 0 \Rightarrow 6a + b = 0$; $\Rightarrow a = -0{,}3$, $b = 1{,}8 \Rightarrow q(x) = -0{,}3x^2 + 1{,}8x$
4.	**$c = 2$; $b = -30a$; $g(x) = -0{,}002x^3 + 0{,}06x^2 + 2$**	$g(0) = 2 \Rightarrow c = 2$; $g'(20) = 0 \Rightarrow 1200a + 40b = 0 \Rightarrow b = -30a$; $g(20) = 10 \Rightarrow 8000a + 400b = 8 \Rightarrow a = -0{,}002$, $b = 0{,}06 \Rightarrow g(x) = -0{,}002x^3 + 0{,}06x^2 + 2$
5.	**$d(x) = 0{,}001x^2 + 4{,}4 - x^4/1\,600\,000$; $x \approx 28{,}3$; Abstand höchstens 4,8 m**	$d(x) = f(x) - g(x) = 0{,}001x^2 + 4{,}4 - x^4/1\,600\,000$; d'(x) $= 0{,}002x - x^3/400\,000 = 0 \Rightarrow x^2 = 800 \Rightarrow x = 20\sqrt{2} \approx 28{,}3$; $d(20\sqrt{2}) = 0{,}8 + 4{,}4 - 0{,}4 = 4{,}8 \Rightarrow$ Abstand höchstens 4,8 m
6.	**$A(x) = x^4 - 6x^3 + 32x$; $x_max \neq 2$**	$A(x) = x \cdot f(x) = x^4 - 6x^3 + 32x$; $A'(x) = 4x^3 - 18x^2 + 32 \Rightarrow A'(2) = 32 - 72 + 32 = -8 \neq 0 \Rightarrow$ bei 2 kein Maximum $\Rightarrow x_max \neq 2$
7.	**$f(0) = 1{,}5$; $b = 2$**	Punkt $(0\|1{,}5) \Rightarrow f(0) = a = 1{,}5$; Punkt $(1\|3) \Rightarrow a \cdot b = 3 \Rightarrow b = 2$